École Polytechnique

Lab Research Project Report

Quantification of generalized measurement incompatibility of mutually unbiased bases

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Academic year 2025/2026

Abstract

In this work we will explore the problem of quantifying the incompatibility of mutually unbiased bases (MUBs) in finite-dimensional Hilbert spaces. We will reproduce the known results on the noise-robustness thresholds for the joint measurability of k MUBs in prime dimensions, and we will present new numerical results obtained by solving semidefinite programs (SDPs) for generalized incompatibility notions where closed-form expressions are not found yet. In producing these numerical results, we will provide a more general framework for quantifying these robustness thresholds by parameterizing the SDPs with arbitrary compatibility hypergraphs, which allows us to explore a wider range of compatibility scenarios beyond full joint measurability. In an attempt to solve the problem analytically, we find analytical lower bounds for the noise-robustness thresholds on these incompatibility scenarios, which match, up to machine precision, the numerical results.

Contents

1 Introduction

Quantum theory and classical physics are fundamentally different even at the level of measurement: quantum measurements are probabilistic and measuring them disturbs, hence, changes the state. Therefore, not every collection of observables can be read out at the same time. Classically, any set of physical quantities can in principle be measured jointly to arbitrary precision, so that a single experiment reveals the value of all of them. In quantum mechanics this is no longer true. There exist sets of measurements that are *incompatible*, in the sense that no single apparatus supplemented only by classical processing of its outcomes can reproduce their statistics. Historically, this feature was first recognised through the non-commutativity of observables, $AB \neq BA$, and the uncertainty relations it entails. From a modern, information-theoretic standpoint, however, incompatibility is much more than a curiosity: it is a necessary ingredient of essentially every genuinely non-classical effect. The violation of Bell inequalities, Einstein–Podolsky–Rosen (EPR) steering, and quantum contextuality all require incompatible measurements, and incompatibility acts as a resource in tasks such as quantum key distribution, quantum state discrimination, and quantum random access codes [5, 2].

The operationally meaningful way to formalise this notion is *joint measurability*. The commutator, while adequate for projective measurements, is too narrow to characterise the most general measurements allowed by quantum mechanics, the positive operator-valued measures (POVMs). Instead, a family of POVMs is said to be jointly measurable, or *compatible*, if there is a single *parent* POVM whose outcomes can be classically post-processed to recover each measurement of the family; if no such parent exists the measurements are *incompatible* (see Definition 4.21). This definition reduces to commutativity for projective measurements but is strictly more permissive in general: there are non-commuting POVMs that nonetheless admit a joint measurement [2].

Among all quantum measurements, those associated with *mutually unbiased bases* (MUBs) are the prototypical candidates for being “maximally incompatible.” Two orthonormal bases of a d-dimensional Hilbert space are mutually unbiased when the overlap of any vector of one basis with any vector of the other has constant modulus, $|\langle b_i|b'_j\rangle|^2 = 1/d$, so that preparing the system in an eigenstate of one basis renders the outcome of a measurement in the other completely random. The eigenbases of the three Pauli operators provide the canonical example in dimension two, and MUBs are commonly regarded as the higher-dimensional generalisation of the Pauli measurements [1].

To turn the binary question “compatible or not” into a quantitative one, one mixes each measurement with white noise. Writing $\Pi_a^{(j)}$ for the projector onto the a-th vector of the j-th basis, the η-noisy MUB measurement has elements

$$M^{\eta}_{a|j} \;=\; \eta\,\Pi_a^{(j)} \;+\; (1-\eta)\,\frac{I}{d}, \qquad j \in [\![1,k]\!],\; a \in [\![1,d]\!], \tag{1}$$

which is the general POVM white-noise mixture $\eta\, M_a + (1-\eta)\,\mathrm{Tr}(M_a)\, I/d$ of (14) specialised to rank-1 projectors – here $\mathrm{Tr}(\Pi_a^{(j)}) = 1$, so the trace factor collapses to unity. One then asks for the largest noise parameter $\eta^\star$ below which the family becomes jointly measurable. This *white-noise robustness* is the figure of merit used throughout this report; crucially, it can be computed by semidefinite programming (SDP) [1, 2]. For the full joint measurability of k MUBs in prime-power dimension d, Designolle, Skrzypczyk, Fröwis and Brunner [1] derived general upper and lower bounds on $\eta^\star(k,d)$ that coincide in several important cases, in particular for two MUBs in any dimension and for the complete set of $k = d+1$ MUBs.

The all-or-nothing notion of full joint measurability is, however, only the coarsest description of how a set of measurements can be (in)compatible. Once three or more measurements are considered, richer *structures* of compatibility appear: a triple can be pairwise compatible yet fail to be jointly measurable as a whole, and one may distinguish, for instance, genuine k-wise incompatibility from weaker forms. Quintino *et al.* [5] organised these possibilities through *compatibility hypergraphs*, in which each hyperedge collects a subset of measurements required to be jointly measurable, and showed how the corresponding robustness can again be cast as an SDP. While the noise robustness of full joint measurability of MUBs is by now well charted, these intermediate scenarios remain largely unexplored quantitatively.

This report addresses precisely that gap. First, we reproduce the known noise-robustness thresholds for the joint measurability of k MUBs by explicitly formulating and solving the corresponding SDPs, recovering the closed-form values of [1]. Second, we provide a general framework in which the robustness SDP is parameterised by an *arbitrary* compatibility hypergraph, allowing us to interpolate between full joint measurability and weaker compatibility scenarios such as the convex combination of pairwise compatibility of k MUBs. Third, we report new numerical values for these generalised incompatibility notions, for which no closed-form expression is yet known. Finally, we derive analytical lower bounds for the corresponding thresholds and show that they match our numerical results up to machine precision, which suggests that the underlying optimal strategy is an equal-probability mixture of all pairwise sub-experiments.

2 Background

Non-commutativity and the rise of joint measurability. The peculiar status of measurement in quantum theory was apparent from the very beginning. In 1925 Heisenberg observed that the product of physical quantities may depend on their order, a remark that Born and Jordan immediately traced back to the matrix nature of observables, $AB \neq BA$ [2]. The ensuing uncertainty relations, of Heisenberg, Kennard and Robertson, tied non-commutativity to the impossibility of preparing states sharply localised with respect to two observables at once. It was later realised that the notion of observables associated to projective measurements is too restrictive to capture all physical measurements, and the notion of a measurement was generalised to POVMs. For this larger class the commutator no longer tells the whole story, and over the past two decades joint measurability has emerged as the central, operationally grounded notion of compatibility, with deep links to Bell nonlocality, EPR steering, contextuality and a range of information-processing tasks [5, 2].

Mutually unbiased bases. Schwinger introduced what we now call mutually unbiased bases in his 1960 *PNAS* paper on unitary operator bases, as a tool for representing pairs of "maximally non-commuting" measurements [6]. Wootters and Fields [7] established that at most $d + 1$ MUBs can exist in dimension d, and gave an explicit construction of complete sets of $d + 1$ MUBs whenever $d = p^r$ is a power of a prime. Whether this bound can be saturated in composite (non-prime-power) dimensions is a long-standing open problem; the smallest open case is $d = 6$, where despite considerable effort no more than three MUBs have ever been found and it is widely conjectured that no fourth one exists. In this report we work exclusively in prime dimensions $d \in \{2, 3, 5, 7, 11\}$, where the Wootters–Fields construction provides explicit complete sets to build on. Beyond their role as extremal incompatible measurements, MUBs are ubiquitous in quantum information processing, appearing in quantum tomography, uncertainty relations, quantum key distribution and entanglement detection [1].

Joint measurability as a semidefinite program. The question of whether a set of POVMs is jointly measurable can be decided by convex optimisation. A key simplification is that the classical post-processing in the definition of joint measurability may, without loss of generality, be taken to be deterministic, so that only finitely many response functions need to be considered [2]. For two binary qubit POVMs, Wolf, Pérez-García and Fernández recast compatibility as the existence of a single positive-semidefinite operator satisfying linear constraints, which is a small feasibility SDP. For a general family $\{M_{a|x}\}$, deciding compatibility amounts to searching for a parent POVM $\{G_\lambda\}$ whose deterministic marginals reproduce every $M_{a|x}$, and this too is an SDP. Replacing the feasibility question by a maximisation over the noise parameter η turns the program into one whose optimal value is exactly the white-noise robustness. Modern interior-point solvers (e.g. CLARABEL, SCS, MOSEK) handle these programs routinely.

Quantifying incompatibility: noise robustness. Just as in entanglement theory one quantifies how far a state is from being separable by adding noise, one quantifies how far a set of measurements is from being compatible.

Mixing each POVM with a fraction of white noise as in Eq. (1) and asking for the critical $\eta^\star$ defines the *(white-)noise robustness*, an incompatibility monotone introduced in this context by Heinosaari, Kiukas and Reitzner and studied as an SDP by Cavalcanti and Skrzypczyk and others [3, 1, 2]. Related quantifiers, such as the incompatibility weight and the (generalised) incompatibility robustness, arise by allowing other noise models, and all share the property of being non-increasing under suitable free operations. For MUBs specifically, Designolle *et al.* [1] obtained, via the dual SDP, a general upper bound on $\eta^\star$ and an analytical lower bound that uses only the unbiasedness relation. These bounds are tight in several families; in particular,

$$\eta^\star(2,d) = \frac{2+\sqrt{d}}{2(\sqrt{d}+1)}, \qquad \eta^\star(d+1,d) = \frac{1}{\sqrt{d+1}}, \tag{2}$$

the former giving $\eta^\star(2,2) = 1/\sqrt{2}$ for the two Pauli MUBs of a qubit. These closed forms populate the reference table of thresholds that we reproduce in Section 6.

Structures of incompatibility. For sets of three or more measurements, compatibility is no longer a single yes/no property but a structured one. Heinosaari, Reitzner and Stano exhibited triples of qubit measurements that are pairwise jointly measurable but not jointly measurable as a whole, and Kunjwal, Heunen and Fritz showed that, for large enough sets, essentially arbitrary joint-measurability structures can be realised in quantum mechanics [2]. Quintino *et al.* [5] formalised this through compatibility hypergraphs $\mathcal{C} = [C_1, \dots, C_N]$, where each hyperedge C_i lists a subset of measurements that must be compatible, and through the notion of *genuine $\mathcal{C}$-incompatibility*: a family is genuinely $\mathcal{C}$-incompatible if it cannot be written as a convex combination of families respecting the structures C_i.

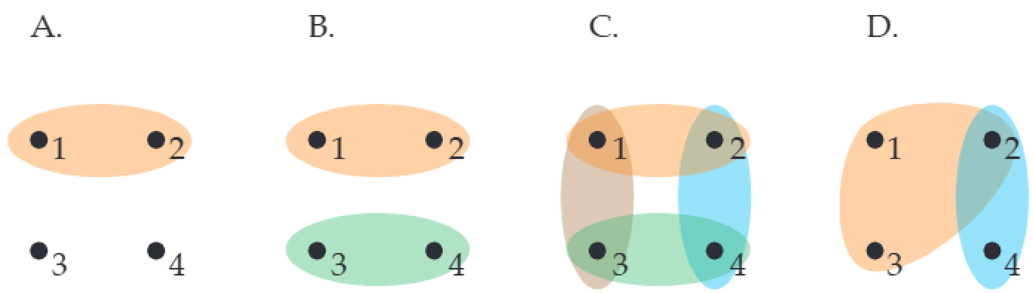

Figure 1: Examples of compatibility hypergraphs on $n = 4$ measurements, reproduced from Fig. 1 of Quintino *et al.* [5]. Each node represents a complete measurement (a POVM); each hyperedge, drawn as a coloured region, groups measurements that are required to be jointly measurable. Vertices not contained in any hyperedge are incompatible with the rest. From left to right: $\mathcal{C}_A = [(1,2)]$, $\mathcal{C}_B = [(1,2),(3,4)]$, $\mathcal{C}_C = [(1,2),(1,3),(2,4),(3,4)]$, $\mathcal{C}_D = [(1,2,3),(2,4)]$. The all-pair hypergraph studied in depth in this work corresponds to C.

The canonical instance is genuine n-wise incompatibility, in which a set of n measurements cannot be decomposed into a mixture of $(n-1)$-wise compatible ones. As in the case of ordinary joint measurability, deciding genuine $\mathcal{C}$-incompatibility and quantifying its noise robustness can be phrased as semidefinite programs [5]. It is exactly this hypergraph-parameterised viewpoint that we adopt and implement in Section 5, in order to quantify incompatibility scenarios for MUBs that lie between full joint measurability and full independence.

3 Outline and goal of the project

The goal of this project is to study, both numerically and analytically, the noise-robustness thresholds that quantify how incompatible mutually unbiased bases (MUBs) are as quantum measurements. Incompatibility is treated here not as a binary property but as a quantitative one: by mixing each MUB measurement with white noise at level $\eta \in [0, 1]$ and asking for the largest η below which the noisy family admits a given compatibility structure, we obtain a sharp figure of merit $\eta^\star$ that can in principle be computed by semidefinite programming. Within this framework, our objectives are the following.

Reproducing the joint-measurability baseline. We first reproduce the known noise-robustness thresholds $\eta^\star(k, d)$ for the full joint measurability of k MUBs in prime dimension d, derived in closed form by Designolle et al. [1] for the special cases $k = 2$ and $k = d + 1$. Concretely, we formulate the corresponding SDP from scratch, implement it in CVXPY, and verify that the solver recovers the analytical values to within 10^{-4} across the full range of (k, d) that is computationally tractable. This step both validates our implementation and produces the reference table against which the generalised scenarios are compared.

Generalising to arbitrary compatibility hypergraphs. Following the hypergraph formalism of Quintino et al. [5], we then design and implement a single SDP whose constraints are parameterised by an arbitrary compatibility hypergraph $\mathcal{C}$. Each hyperedge of $\mathcal{C}$ specifies a subset of measurements that must admit a common parent POVM, while the remaining measurements in each branch are left as independent POVMs; the SDP searches for the largest η such that the noisy MUB family decomposes as a convex mixture of such branches. The trivial hypergraph $\mathcal{C} = \{(0, 1, \ldots, k-1)\}$ recovers full joint measurability as a special case, so the standard problem becomes the simplest instance of a much broader family of compatibility scenarios that our code handles uniformly.

Producing new numerical thresholds for the all-pairs hypergraph. Using this general framework, we compute new noise-robustness thresholds for the all-pairs compatibility hypergraph, in which the k measurements are required to be only pairwise—rather than fully—jointly measurable, in a convex-mixture sense. Because this SDP scales only as $O(k^3 d^2)$ in the number of matrix variables, rather than as $O(d^k)$ for the full-JM SDP, we are able to populate a substantially larger range of (k, d) values than is feasible for the joint-measurability case, up to $k = 9$ in dimension $d = 11$.

Deriving and certifying an analytical lower bound. Finally, we exploit the symmetry of the MUB family to construct an explicit feasible point of the all-pairs SDP, given by the equal-weight mixture of all $\binom{k}{2}$ pairwise strategies. Combining this construction with the affineness of the white-noise map and the closed-form pairwise threshold $\eta^\star(2, d)$ from [1], we obtain the analytical lower bound

$$\eta^\star_{\text{pair}}(k, d) \;\geq\; 1 - \frac{\sqrt{d}}{k(\sqrt{d}+1)}. \tag{3}$$

This bound is monotonically increasing in k, in sharp contrast to the full-JM threshold which decreases in k, and it matches our numerical SDP values to solver precision in every case we tested. The agreement strongly suggests that the equal-weight pairwise mixture is in fact optimal for this hypergraph, and that the bound is tight.

4 Definitions and Concepts

4.1 Hermitian operators

The definitions in this subsection are standard and follow the presentation of [4]; we include them here for self-containedness and to fix the notation used in the rest of the report.

Definition 4.1 (Pre-Hilbert space)**.** A pair $(\mathcal{H}, \langle\ ,\ \rangle)$ composed of complex vector space E and an inner product $\langle\ ,\ \rangle$ on E is called a complex inner product space, or a complex pre-Hilbert space.

Definition 4.2 (Hermitian space)**.** A finite-dimensional pre-Hilbert space is called a Hermitian space.

Definition 4.3 (Cauchy sequence)**.** A sequence $(\psi_n)_{n\in\mathbb{N}}$ in a pre-Hilbert space $\mathcal{H}$ is called a Cauchy sequence if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $m, n > N$, we have $\|\psi_n - \psi_m\| < \epsilon$.

Definition 4.4 (Complete space)**.** A pre-Hilbert space is called complete if every Cauchy sequence converges.

Definition 4.5 (Hilbert space)**.** A complete pre-Hilbert space is called a Hilbert space.

Definition 4.6 (Adjoint of an operator in a Hilbert space)**.** Let $\mathcal{H}$ be a complex Hilbert space. For any linear operator $A : \mathcal{H} \to \mathcal{H}$, there exists a unique linear operator $A^\dagger : \mathcal{H} \to \mathcal{H}$ such that

$$\langle\phi|A\psi\rangle = \langle A^\dagger\phi|\psi\rangle \tag{4}$$

for all $\phi, \psi \in \mathcal{H}$. The operator $A^\dagger$ is called the adjoint of A.

Definition 4.7 (Hermitian operator)**.** Let $\mathcal{H}$ be a complex Hilbert space. A linear operator $A : \mathcal{H} \to \mathcal{H}$ is called *Hermitian* (or *self-adjoint*) if it equals its own adjoint,

$$A \;=\; A^\dagger, \tag{5}$$

or, equivalently, $\langle\phi|A\psi\rangle = \langle A\phi|\psi\rangle$ for all $\phi, \psi \in \mathcal{H}$.

In finite dimension $\mathcal{H} = \mathbb{C}^d$, this reduces to the matrix condition $A_{ij} = \overline{A_{ji}}$. Hermitian operators have real eigenvalues and an orthonormal eigenbasis, which is why they model physical observables in quantum mechanics.
Also note that a Hermitian space (finite-dimenional pre-Hilbert space) is always complete, and thus it is a Hilbert space. In this study we are only concerned with finite-dimensional Hilbert spaces. While we will use the term "Hilbert space", it is important to remember that the spaces we are working with are also Hermitian spaces.

Definition 4.8 (Positive semidefinite operator)**.** Let $\mathcal{H}$ be a complex Hilbert space. A linear operator $A : \mathcal{H} \to \mathcal{H}$ is called positive semidefinite if for all $\psi \in \mathcal{H}$, we have $\langle\psi|A\psi\rangle \geq 0$.

4.2 Quantum states, density operators and quantum measurements

A quantum state is a mathematical object that fully describes the quantum mechanical state of a physical system. It can be represented by a state vector in a complex Hilbert space or by a density operator.

Definition 4.9 (Quantum Bit (Qubit))**.** A qubit is a linear combination of the (orthonormal) computational basis states $|0\rangle$ and $|1\rangle$ in a two-dimensional complex Hilbert space $\mathcal{H} = \mathbb{C}^2$. A general state of a qubit can be written as

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \tag{6}$$

where $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$. Upon measuring the qubit in the computational basis, the outcome will be $|0\rangle$ with probability $|\alpha|^2$ and $|1\rangle$ with probability $|\beta|^2$.

Definition 4.10 (Quantum state)**.** A quantum state can be described by a state vector $|\psi\rangle$ in a complex Hilbert space $\mathcal{H}$, where $|\psi\rangle$ is a unit vector (i.e., $\langle\psi|\psi\rangle = 1$). The state vector represents the pure state of a quantum system.

Definition 4.11 (Pure state)**.** A pure state is a quantum state that can be represented by a state vector $|\psi\rangle$ in a complex Hilbert space. It is called pure because it cannot be expressed as a convex combination of other states. A pure state corresponds to a rank-1 projector in the density operator formalism, i.e., $\rho = |\psi\rangle\langle\psi|$.

Definition 4.12 (Density operator)**.** A density operator (or density matrix) ρ is a positive semidefinite operator on a complex Hilbert space $\mathcal{H}$ with trace equal to 1, i.e., $\rho \geq 0$ and $\mathrm{Tr}(\rho) = 1$. Density operators can represent both pure states (when ρ is a rank-1 projector) and mixed states (when ρ is a convex combination of pure states). For a quantum state described by the set $\{p_i, |\psi_i\rangle\}$, where p_i are probabilities and $|\psi_i\rangle$ are pure states, the corresponding density operator is given by

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| . \tag{7}$$

4.3 Mutually unbiased basis (MUBs)

Definition 4.13 (Basis)**.** Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space, with $dim(\mathcal{H}) = d$. A set of vectors $\mathcal{B} = \{|b_i\rangle\}_{i\in I}$ is called a basis of $\mathcal{H}$ if every vector $|\psi\rangle \in \mathcal{H}$ can be uniquely expressed as a linear combination of the vectors in $\mathcal{B}$, i.e., there exist unique coefficients $c_i \in \mathbb{C}$ such that

$$|\psi\rangle = \sum_{i\in I} c_i |b_i\rangle . \tag{8}$$

Definition 4.14 (Orthonormal basis)**.** A basis $\mathcal{B} = \{|b_i\rangle\}_{i=1}^d$ of a complex Hilbert space $\mathcal{H}$ is called orthonormal if for all $i, j \in \{1, \ldots, d\}$, we have

$$\langle b_i|b_j\rangle = \delta_{ij}, \tag{9}$$

where δ_{ij} is the Kronecker delta function, which equals 1 if $i = j$ and 0 otherwise.

Definition 4.15 (Mutually unbiased bases (MUBs))**.** Let $\mathcal{H}$ be a complex Hilbert space of dimension d. Two orthonormal bases $\mathcal{B} = \{|b_i\rangle\}_{i=1}^d$ and $\mathcal{B}' = \{|b'_j\rangle\}_{j=1}^d$ of $\mathcal{H}$ are called mutually unbiased if for all $i, j \in \{1, \ldots, d\}$, we have

$$|\langle b_i|b'_j\rangle|^2 = \frac{1}{d}. \tag{10}$$

A set of orthonormal bases $\{\mathcal{B}_1, \ldots, \mathcal{B}_k\}$ is called mutually unbiased if every pair of distinct bases in the set is mutually unbiased.

4.4 Quantum Measurements

A quantum measurement is the formal device by which one extracts classical information from a quantum state. Each measurement is specified by a finite set of *outcomes* $\{a\}_{a=1}^n$ together with an operator on the Hilbert space $\mathcal{H}$ for each outcome; the probability of each outcome is then read off from the state via the Born rule below. We distinguish two levels of generality, projective measurements and the strictly larger class of POVMs.

Definition 4.16 (Projective measurement (PVM))**.** A *projective measurement* on a complex Hilbert space $\mathcal{H}$ of dimension d is a finite family $\{\Pi_a\}_{a=1}^n$ of orthogonal projectors on $\mathcal{H}$,

$$\Pi_a^\dagger = \Pi_a, \qquad \Pi_a\Pi_b = \delta_{ab}\,\Pi_a, \qquad \sum_{a=1}^n \Pi_a = I. \tag{11}$$

By the spectral theorem, projective measurements are in one-to-one correspondence with self-adjoint observables $A = \sum_a \lambda_a \Pi_a$ on $\mathcal{H}$, the λ_a being the eigenvalues attached to each outcome.

Definition 4.17 (Positive operator-valued measure (POVM))**.** A *positive operator-valued measure* (POVM) on a complex Hilbert space $\mathcal{H}$ with outcome set $\{1, \dots, n\}$ is a family $\{M_a\}_{a=1}^n$ of Hermitian operators satisfying

$$M_a \;\geq\; 0 \quad \forall\, a, \qquad \sum_{a=1}^{n} M_a \;=\; I. \tag{12}$$

Every projective measurement is a POVM (the Π_a are positive, Hermitian, and sum to the identity), but the converse fails: a POVM element M_a may have rank greater than one and the M_a need not mutually commute. POVMs are the most general notion of measurement allowed by quantum mechanics; by Naimark's theorem every POVM arises as a projective measurement on an enlarged Hilbert space, with the extra degrees of freedom subsequently traced out.

Definition 4.18 (Born rule)**.** If the system is prepared in a state described by the density operator ρ and a POVM $\{M_a\}$ is performed, the probability of observing outcome a is

$$P(a \mid \rho) \;=\; \mathrm{Tr}(\rho\, M_a). \tag{13}$$

Positivity $M_a \geq 0$ together with the resolution of the identity $\sum_a M_a = I$ guarantees that $\{P(a \mid \rho)\}_{a=1}^n$ is a valid probability distribution on the outcome set for every density operator ρ.

Indexed families of POVMs. In what follows we shall study not a single POVM but *collections* of POVMs sharing the same Hilbert space and outcome set, indexed by a classical measurement *setting* $x \in \{1, \dots, m\}$. We adopt the standard notation $M^{(x)} = \{M_{a|x}\}_{a=1}^n$ for the POVM associated with setting x, so that each $M_{a|x}$ is a Hermitian operator on $\mathcal{H}$ satisfying (12) for every fixed x, and the Born rule extends naturally to $P(a \mid x, \rho) = \mathrm{Tr}(\rho\, M_{a|x})$. The central question of this report—developed in the next subsection and formalised as a semidefinite program in Section 5.1—is to what extent such a collection $\{M^{(x)}\}_x$ can be realised *simultaneously* by a single physical measurement.

Definition 4.19 (White-noise mixture of a POVM)**.** Given a POVM $\{M_a\}$ on $\mathcal{H} = \mathbb{C}^d$ and a noise parameter $\eta \in [0, 1]$, the corresponding *η-noisy* POVM has elements

$$M_a^{\eta} \;=\; \eta\, M_a \;+\; (1-\eta)\, \mathrm{Tr}(M_a)\, \frac{I}{d}. \tag{14}$$

At $\eta = 1$ the original sharp measurement is recovered; at $\eta = 0$ every outcome operator becomes proportional to the maximally mixed state I/d, so the outcome distribution $P(a \mid \rho)$ no longer depends on ρ and the measurement is uninformative. The parameter η will serve as the quantitative figure of merit of measurement incompatibility throughout the rest of the report.

Lemma 4.20 (Affineness)**.** *The map $\eta \mapsto M^{\eta} = \eta M + (1-\eta)\, \mathrm{Tr}(M)\, \frac{I}{d}$ is affine, so for any probabilities $\{q_b\}$ and noise levels $\{\eta_b\}$,*

$$\sum_b q_b\, M^{\eta_b} \;=\; M^{\sum_b q_b \eta_b}.$$

4.5 Joint Measurability and Incompatibility

Definition 4.21 (Joint measurability)**.** A set of quantum measurements (POVMs) $\{M_{a|x}\}$ is said to be jointly measurable if there exists a single measurement (POVM) $\{G_\lambda\}$ such that the outcomes of G can be post-processed to reproduce the statistics of each $M_{a|x}$. That is, there exists a set of conditional probabilities

$p(a|x, \lambda)$ such that for all x and a, we have

$$M_{a|x} = \sum_{\lambda} p(a|x, \lambda) G_{\lambda}, \tag{15}$$

In this case the POVM $\{G_\lambda\}$ is called a joint or parent measurement of the set $\{M_{a|x}\}$.

A set of quantum measurements is **incompatible** if it is not jointly measurable.

4.6 Semi Definite Programming (SDP)

The feasibility and noise-robustness of measurement compatibility can be modeled as an SDP and computationally solved by convex optimization libraries, for example in Python or MatLAB. For this reason, we will briefly review the definition of an SDP and some prerequisite concepts, and in the next section we will model our systems as SDPs.

Definition 4.22 (Positive semidefiniteness of a Hermitian operator)**.** A Hermitian operator A is said to be positive semidefinite if for all vectors $|\psi\rangle$ in the Hilbert space, we have

$$\langle\psi|A|\psi\rangle \geq 0. \tag{16}$$

We will denote this by $A \geq 0$.

Note that we can further extend this (non-total) ordering of Hermitian operators like so

$$A \geq B \quad \text{if and only if} \quad A - B \geq 0. \tag{17}$$

Definition 4.23 (Semidefinite Program)**.** A semidefinite program is an optimization problem of the form

$$\begin{aligned} \text{minimize} \quad & \mathrm{Tr}(CX) \\ \text{subject to} \quad & \mathrm{Tr}(A_i X) = b_i, \quad i = 1, \dots, m, \\ & X \geq 0, \end{aligned} \tag{18}$$

where X is the variable being optimized over, C and A_i are given Hermitian matrices, and b_i are given scalars.

5 Joint Measurability SDP

In this section, we will explore convex optimization programs for identifying jointly measurable sets of quantum measurements and quantifying their incompatibility by solving these programs for minimum noise on noisy measurements. We aim to derive and implement a generalization of this program in order to obtain more numerical results on the genuine k-wise incompatibility problem introduced by Quintino et al. [5].

5.1 A first example: two noisy Pauli measurements on a qubit

To make the framework concrete, we begin with the smallest non-trivial case: two binary-outcome POVMs on a single qubit, i.e. $d = 2$, $m = 2$ settings, $n_a = 2$ outcomes per setting. Take the eigenprojectors of the Pauli observables σ_X and σ_Z,

$$\Pi_{\pm}^{(X)} = \frac{I \pm \sigma_X}{2}, \qquad \Pi_{\pm}^{(Z)} = \frac{I \pm \sigma_Z}{2}, \tag{19}$$

and mix them with white noise at level $\eta \in [0, 1]$,

$$M_{a|x}^{\eta} = \eta\, \Pi_a^{(x)} + (1 - \eta)\, \frac{I}{2}, \qquad x \in \{X, Z\},\ \ a \in \{+, -\}. \tag{20}$$

By construction each $M_{a|x}^{\eta}$ is positive semidefinite and $\sum_a M_{a|x}^{\eta} = I$, so $\{M_{\cdot|x}^{\eta}\}$ is a POVM for every setting x and every $\eta \in [0, 1]$.

By the definition of joint measurability given above, the family $\{M_{a|x}^{\eta}\}_{x\in\{X,Z\},\, a\in\{+,-\}}$ is jointly measurable iff there exists a single parent POVM $\{G_\lambda\}_{\lambda\in\{+,-\}^2}$ on $\mathbb{C}^2$ whose marginals recover M^η at both settings $x = X$ and $x = Z$. Asking this question *directly* for fixed η yields the semidefinite feasibility program in the four $2{\times}2$ Hermitian matrices $G_{++}, G_{+-}, G_{-+}, G_{--}$:

$$\begin{aligned}
\text{find}\ \ & G_{\lambda_X\lambda_Z} \in \mathbb{C}^{2\times 2} \quad \forall\, (\lambda_X, \lambda_Z) \in \{+, -\}^2, \\
\text{s.t.}\ \ & G_{\lambda_X\lambda_Z} \geq 0 \quad \forall\, (\lambda_X, \lambda_Z), \\
& \sum_{\lambda_X, \lambda_Z} G_{\lambda_X\lambda_Z} = I, \\
& \sum_{\lambda_Z\in\{+,-\}} G_{a\,\lambda_Z} = M_{a|X}^{\eta} \qquad \forall\, a \in \{+, -\}, \\
& \sum_{\lambda_X\in\{+,-\}} G_{\lambda_X\, a} = M_{a|Z}^{\eta} \qquad \forall\, a \in \{+, -\}.
\end{aligned} \tag{21}$$

The first two lines state that $\{G_\lambda\}$ is a valid four-outcome POVM; the last two state that summing out one outcome of the parent measurement recovers the corresponding noisy Pauli measurement. The feasible set is the intersection of an affine subspace with the positive-semidefinite cone, hence convex, and the problem is solved in practice by any modern interior-point SDP solver (CLARABEL, SCS, MOSEK).

Result. For fixed η, the program (21) is feasible iff

$$\eta \leq \eta^\star(2, 2) = \frac{1}{\sqrt{2}} \approx 0.7071, \tag{22}$$

the well-known incompatibility threshold for two mutually unbiased Pauli observables on a qubit; this is the $k = 2$, $d = 2$ entry of Table I of [1]. Below the threshold the noisy measurements admit a joint POVM and are therefore compatible; above it no joint POVM exists and they are genuinely incompatible.

Noise Robustness SDP To move from a feasibility program to an optimization program, we shift the goal from finding a parent POVM to finding the biggest η (minimum noise) such that the noisy measurements are jointly measurable. The resulting optimization program is

$$\begin{aligned} \text{maximize} \;\; & \eta \;\in\; [0,1] \\ \text{s.t.} \;\; & G_{\lambda_X \lambda_Z} \;\in\; \mathbb{C}^{2\times 2} \quad \forall\,(\lambda_X, \lambda_Z) \in \{+,-\}^2, \\ & G_{\lambda_X \lambda_Z} \;\geq\; 0 \quad \forall\,(\lambda_X, \lambda_Z), \\ & \sum_{\lambda_X, \lambda_Z} G_{\lambda_X \lambda_Z} \;=\; I, \\ & \sum_{\lambda_Z \in \{+,-\}} G_{a\,\lambda_Z} \;=\; M^{\eta}_{a|X} \qquad \forall\, a \in \{+,-\}, \\ & \sum_{\lambda_X \in \{+,-\}} G_{\lambda_X\,a} \;=\; M^{\eta}_{a|Z} \qquad \forall\, a \in \{+,-\}. \end{aligned} \tag{23}$$

whose optimal value is exactly the white-noise robustness threshold (22). The right-hand side of the marginal constraints in (21) now depends *affinely* on the decision variable η, so (23) is still an SDP; in fact, it is the standard linear-objective semidefinite program. This is the template that we will generalise in the next subsections to m noisy MUBs in arbitrary dimension d, and ultimately to arbitrary compatibility hypergraphs.

5.2 Generalization of the robustness SDP to k MUB measurements with m outcomes

To generalize the SDP to k noisy MUB measurements with m outcomes in dimension d, we replace the binary settings and outcomes of the previous example with k settings and m outcomes per setting, and we replace the noisy Pauli measurements with noisy MUB measurements defined as

$$M^{\eta}_{a|j} \;=\; \eta\,\Pi^{(j)}_a \;+\; (1-\eta)\,\frac{I}{d}, \qquad j \in [\![1,k]\!], \;\; a \in [\![1,m]\!], \tag{24}$$

where $\Pi^{(j)}_a$ are the projectors onto the a-th vector of the j-th MUB. The parent POVM $\{G_\lambda\}$ now has m^k outcomes, indexed by k-tuples $\lambda_1 \cdots \lambda_k$ of outcomes, one for each setting. The marginal constraints now require summing out all outcomes of the parent POVM except the one corresponding to the setting of interest, which is done by summing over all k-tuples $\lambda_1 \cdots \lambda_k$ such that $\lambda_j = a$ for the setting j and outcome a of interest.

$$\begin{aligned} \text{maximize} \;\; & \eta \;\in\; [0,1] \\ \text{s.t.} \;\; & G_{\lambda_1 \cdots \lambda_k} \;\in\; \mathbb{C}^{d\times d} \quad \forall\,(\lambda_1, \ldots, \lambda_k) \in [\![1,m]\!]^k, \\ & G_{\lambda_1 \cdots \lambda_k} \;\geq\; 0 \quad \forall\,(\lambda_1, \ldots, \lambda_k), \\ & \sum_{\lambda_1, \ldots, \lambda_k} G_{\lambda_1 \cdots \lambda_k} \;=\; I, \\ & \sum_{\substack{\lambda_1, \ldots, \lambda_k \\ \lambda_j = a}} G_{\lambda_1 \cdots \lambda_k} \;=\; M^{\eta}_{a|j} \qquad \forall\, j \in [\![1,k]\!],\; \forall\, a \in [\![1,m]\!]. \end{aligned} \tag{25}$$

This optimization program can be numerically solved by passing the compatibility hypergraph produced by `hypergraph_compatible(k)` (Listing 3) to the `robustness_hypergraph` routine of Appendix A.1 (Listing 1), which builds the convex program above with CVXPY and delegates the actual optimisation to a standard interior-point SDP solver such as CLARABEL, SCS, or MOSEK. The single hyperedge $\{(0, 1, \ldots, k-1)\}$ returned by `hypergraph_compatible` collapses the decomposition to one branch in which all k noisy measurements are forced to share a common parent POVM, so that (25) reduces exactly to the standard k-wise joint-measurability robustness SDP whose optimal value populates Table 1.

5.3 Generalization of the robustness SDP to pairwise jointly measurable hypergraphs

Now that we have established certain robustness bounds for the joint measurability of k MUBs, we can explore a more general notion of compatibility, which is the k-wise incompatibility of k MUBs. More specifically, we are interested in finding a probabilistic mixture of strategies, in each of which only one pair of the k measurements is jointly measured and the remaining k-2 are measured independently that would yield a valid and identically distributed measurement for each of the k MUBs, and we want to find the noise-robustness threshold for this compatibility scenario. This is a natural generalization of the full joint measurability scenario, and it is quite clear, and proved in the following, that this specific program will yield higher robustness thresholds.

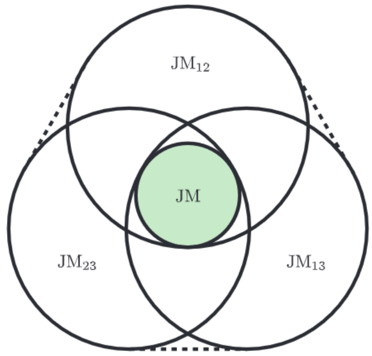

Figure 2: Convex-combination view of joint measurability for three noisy measurements, reproduced from Quintino *et al.* [5]: the family is jointly measurable iff it lies in the convex hull (shaded) of single-branch subfamilies. JM illustrates the set of fully jointly measurable families of MUBs, while JM_{12}, JM_{13}, JM_{23} represent the sets of families in which the indicated pair is jointly measurable, with no constraint on the third measurement.

First, let us generalize the notion of genuine incompatibility from [5] to genuine k-wise incompatibility:
A set of k measurements is genuinely k-wise incompatible when it cannot be written as a convex combination of strategies of $k-1$ measurements where 2 are jointly measured and the rest are measured individually. By this, we mean choosing a measurement strategy in which we probabilistically choose a pair of measurements $i \neq j$ with $i, j \in [\![1, k]\!]$ and perform a joint measurement for that pair while measuring the rest individually.
More formally, we have a probability matrix P_{ij} with $i, j \in [\![1, k]\!]$ and $i \neq j$, such that $\sum_{i<j} P_{ij} = 1$, and we have a set of parent POVMs $\{G_\lambda^{(ij)}\}$ for each pair of measurements $i \neq j$, such that the marginals of $G_\lambda^{(ij)}$ recover the noisy MUB measurements at settings i and j.
Geometrically (Figure 2), the feasibility of this SDP amounts to asking whether the noisy family $\{M^\eta_{a|j}\}$ lies in the convex hull of the single-branch jointly-measurable families. Figure 2 illustrates why the noise-robustness threshold for full joint measurability is lower than the noise-robustness threshold for pairwise joint measurability: the set of fully jointly measurable families of MUBs (JM) is a strict subset of the convex hull of the pairwise jointly measurable families $(JM_{12}, JM_{13}, JM_{23})$.
The baseline program becomes:

$$
\begin{aligned}
\text{find} \qquad & J_{a|x}^{(ij)},\ p_{ij},\ E_\lambda^{(ij)} \text{ for all } i, j \in [\![1, k]\!],\ i < j \\
\text{such that} \qquad & E_\lambda^{(ij)} \geq 0, \quad p_{ij} \geq 0, \\
& M_{a|x} = \sum_{i<j} p_{ij} J_{a|x}^{(ij)}, \\
& J_{a|x}^{(ij)} \geq 0,\ \forall a, x;\ \sum_a J_{a|x}^{(ij)} = \mathbb{1},\ \forall x, \\
& J_{a|x}^{(ij)} = \sum_\lambda D_\lambda(a|x)\, E_\lambda^{(ij)} \text{ for } x \in \{i, j\}.
\end{aligned}
$$

where $D_\lambda(a|x)$ is the deterministic response function for the parent POVM $G_\lambda^{(ij)}$, and the constraints ensure that each $J_{a|x}^{(ij)}$ is a valid measurement strategy that recovers the noisy MUB measurements at settings i and j

as marginals, and that the overall decomposition of $M_{a|x}$ into these pairwise strategies is valid.
Notice that we cannot yet use convex optimization to solve this program, since the constraints are **not linear** due to the product of **variables** p_{ij} and $J^{(ij)}_{a|x}$ in the decomposition of $M_{a|x}$. However, we can easily fix this by integrating the probability matrix p into the Joint measurement term J:

$$\begin{aligned}
&\text{find} && J^{(ij)}_{a|x},\ p_{ij},\ E^{(ij)}_\lambda \text{ for all } i,j \in [\![1,k]\!],\ i<j \\
&\text{such that} && E^{(ij)}_\lambda \geq 0, \quad p_{ij} \geq 0, \\
& && M_{a|x} = \sum_{i<j} J^{(ij)}_{a|x}, \\
& && J^{(ij)}_{a|x} \geq 0,\ \forall a,x;\ \sum_a J^{(ij)}_{a|x} = p_{ij}\mathbb{1},\ \forall x, \\
& && J^{(ij)}_{a|x} = \sum_\lambda D_\lambda(a|x)\, E^{(ij)}_\lambda \text{ for } x \in \{i,j\}.
\end{aligned} \tag{26}$$

In order to quantify the noise-robustness threshold η^* for the k-wise compatibility scenario, we add the maximization constraint and model the following constraints:

$$\begin{aligned}
\eta M_{a|x} + (1-\eta)\,\mathrm{Tr}(M_{a|x})\frac{\mathbb{1}}{d} &= J^k_{a|x}, \\
J^k_{a|x} &= \sum_{i<j} J^{(ij)}_{a|x}.
\end{aligned} \tag{27}$$

Since, $J^{(ij)}_{a|x} = \sum_\lambda D_\lambda(a|x)\, E^{(ij)}_\lambda$, $E^{(ij)}_\lambda$ is proportional to a POVM and $\sum_{i<j} E^{(ij)}_\lambda$ is a POVM, we can further simplify the program as follows:

$$\begin{aligned}
&\text{maximize} && \eta \in [0,1] \\
&\text{such that} && E^{(ij)}_\lambda \geq 0, \sum_\lambda E^{(ij)}_\lambda = \frac{\mathbb{1}}{d}\sum_\lambda Tr(E^{(ij)}_\lambda),\ \forall i,j \in [\![1,k]\!],\ i<j, \\
& && \frac{1}{d}\sum_\lambda \sum_{i<j} Tr(E^{(ij)}_\lambda) = 1 \\
& && \eta M_{a|x} + (1-\eta)Tr(M_{a|x})\frac{\mathbb{1}}{d} \geq \sum_\lambda D_\lambda(a|x) \sum_{i\neq x} E^{(ix)}_\lambda\ \forall a,x.
\end{aligned}$$

Where we extend E such that $E^{(ij)}_\lambda = E^{(ji)}_\lambda$ for $i>j$.
This state of the program is easy to solve by a convex optimization engine as well as to write as an SDP for general k, which has been done in the `robustness_hypergraph` routine of Appendix A.1 (Listing 1), which takes as input the compatibility hypergraph produced by `hypergraph_pair(k)`, and builds the convex program above with CVXPY, delegating the actual optimization to a standard interior-point SDP solver such as CLARABEL, SCS, or MOSEK.

We then identify a natural lower bound η^* for the noise-robustness threshold of this $(k-1)$-wise compatibility scenario, which is obtained by an equal probability mixture of all $\binom{k}{2}$ pairwise strategies, i.e. $p_{ij} = \frac{1}{\binom{k}{2}}$

for all $i < j$. To compute η^*, we must first acknowledge the noise-robustness threshold for the pairwise compatibility of measurements i and j, from Designolle et al. [1], which is a tight bound for the particular case of $k = 2$:

$$\eta^\star(2, d) = \frac{2 + \sqrt{d}}{2(\sqrt{d} + 1)}. \tag{28}$$

Call a branch (j, ℓ) **active for setting** x if $x \in \{j, \ell\}$, and **inactive** otherwise. In an active branch, setting x is presented at noise level $\eta^\star(2, d)$; in an inactive branch, at noise level 1.
Now, let us fix a measurement setting x. By MUB symmetry, x belongs to exactly $k - 1$ active branches and to $\binom{k}{2} - (k - 1)$ inactive ones. By the affineness lemma the mixture equals $M_{a|x}^{\eta_{\text{eff}}}$ with:

$$\eta_{\text{eff}} = \frac{1}{\binom{k}{2}} \left(\binom{k-1}{1} \eta^\star(2, d) + \left(\binom{k}{2} - \binom{k-1}{1} \right) \cdot 1 \right) \tag{29}$$

Symmetry guarantees that the same number of active branches cover every setting x, hence the mixture reconstructs the noisy MUB family uniformly. By simplifying, we get:

$$\begin{aligned} \eta_{\text{eff}} &= \frac{2}{k(k-1)}((k-1)\eta^*(2, d) + \frac{k(k-1)}{2} - (k-1)) \\ &= \frac{1}{k}(2\eta^*(2, d) + k - 2) \\ &= 1 + \frac{1}{k}(\frac{2 + \sqrt{d}}{\sqrt{d} + 1} - 2) \\ &= 1 - \frac{\sqrt{d}}{k(\sqrt{d} + 1)} \end{aligned} \tag{30}$$

In the results section, we will see that, for the cases we compute numerical values, this lower bound is certified by the SDP, which leads us to believe that the optimal strategy in the general case is to choose each pairwise strategy with equal probability. Another thing to note about this lower bound (which holds for our computed values) is that it is **decreasing** in d, just like the noise-robustness threshold for Joint measurability, however it is **increasing** in k, unlike the threshold for Joint Measurability.
Let us develop on the latter conclusion in more detail: Adding a measurement to the full-JM hypergraph enlarges the parent POVM by a factor of d and imposes one more marginal constraint, making the joint POVM strictly harder to find. Adding a measurement to the pairwise hypergraph, on the other hand, only adds new branches: the decomposition gains k more sub-strategies, each of which has to honour the marginals of a single pair while leaving the remaining $k - 2$ measurements as arbitrary POVMs. The "responsibility" of any fixed measurement is consequently distributed across $k - 1$ branches out of a total of $\binom{k}{2}$, a vanishing fraction $2/k$ that becomes negligible as k grows. Quantitatively, this is exactly the hierarchy of compatibility hypergraphs introduced by Quintino *et al.* [5]: finer hypergraphs (with more, smaller hyperedges) yield monotonically larger noise tolerances, and the all-pairs hypergraph – being one of the finest non-trivial structures – yields one of the loosest possible relaxations of full joint measurability.

6 Results

Numerical setup. All SDPs were formulated with CVXPY and solved using CLARABEL for the small cases $d^k \leq 100$ and SCS with $\epsilon = 10^{-7}$, max_iters $= 10^5$ for the larger ones, on a standard laptop CPU. Cells with $d^k > 3000$ were omitted from Table 1 as the runtime exceeds minutes; Table 2 has no such cap since the pairwise SDP scales as $\mathcal{O}(k^3 d^2)$. Every closed-form entry in Tables 1 and 2 agreed with the analytical expression to within 10^{-4}, and the solver reported `optimal` (never `optimal_inaccurate`) status for all reported cells.

In this section, we present the numerical results obtained. In the first table, we reproduce the noise-robustness thresholds for the joint measurability of k MUBs, while in the second table, we study the values obtained for the generalized k-wise incompatibility (for prime dimensions), that we obtained with our new generalized program. Note that we are able to expand the second table much more than the first one, which at first could seem counterintuitive since the pairwise hypergraph is generating more constraints on the optimization problem.
However, the parent POVM of a Joint measurement has d^k outcomes, hence in the full joint measurability scenario, we are optimizing over d^k positive semidefinite matrices of size $d \times d$, a number that grows *exponentially* with k. In contrast, the pairwise hypergraph contains $\binom{k}{2}$ branches, each of which carries a parent POVM with only d^2 outcomes plus $(k-2)$ independent single-measurement POVMs; the total variable count is therefore $\binom{k}{2}\big(d^2 + (k-2)\,d\big) = \mathcal{O}(k^3 d + k^2 d^2)$, which grows only *cubically* with k. The same gap shows up in the marginal constraints, where the full-Jointly Measurable SDP sums d^{k-1} matrices per equality while the pairwise SDP sums only d matrices per branch. Concretely, for $(k = 8,\ d = 7)$ the full-Jointly Measurable SDP would require $7^8 \approx 5.76 \cdot 10^6$ matrix variables (intractable on any desktop), while the pairwise SDP requires $\binom{8}{2}(7^2 + 6 \cdot 7) = 2{,}548$ matrix variables which can be solved in seconds, even on CPU. For this reason, we were able to populate table 2 up to $k = 11$ and $d = 9$ in less than 10 minutes, while having to omit even a few seemingly small entries from table 1, such as $(k, d) = (5, 5)$.

Table 1: Numerical values for measurement compatibility: noise-robustness thresholds $\eta^\star(k, d)$ for the joint measurability of k MUBs in prime dimension d. Closed-form entries are taken from [1] (or, for the complete set $k = d+1$, from its Theorem 1, which gives $\eta^\star(d+1, d) = 1/\sqrt{d+1}$); cells marked "X" have been omitted due to computational complexity; the closed forms can be found in [1]. "—" marks cells with $k > d + 1$, for which fewer than k MUBs exist.

	d			
k	2	3	5	7
2	$\frac{1}{\sqrt{2}} \approx 0.7071$	$\frac{1+\sqrt{3}}{4} \approx 0.6830$	$\frac{3+\sqrt{5}}{8} \approx 0.6545$	$\frac{5+\sqrt{7}}{12} \approx 0.6371$
3	$\frac{1}{\sqrt{3}} \approx 0.5774$	$\frac{\cos(\pi/18)}{\sqrt{3}} \approx 0.5686$	$\frac{1+\sqrt{5}}{6} \approx 0.5393$	0.5101
4	—	$\frac{1}{2} = 0.5000$	0.4615	0.4436
5	—	—	X	X
6	—	—	$\frac{1}{\sqrt{6}} \approx 0.4082$	X
7	—	—	—	X
8	—	—	—	$\frac{1}{\sqrt{8}} \approx 0.3536$

Table 2: Noise-robustness thresholds for the all-pairs compatibility hypergraph on k MUBs in prime dimension d, computed by the SDP of Section 5.3 with hypergraph `hypergraph_pair(k)`. The hyperedges are all $\binom{k}{2}$ pairs of settings; below the threshold the noisy MUB family admits a convex decomposition into pairwise-jointly-measurable sub-experiments and is therefore not genuinely incompatible with respect to this structure. All entries coincide with the closed form $\eta^\star = 1 - \sqrt{d}/(k(\sqrt{d}+1))$ derived in Eq. 30, confirming optimality of the equal-weight pairwise mixture to solver precision. "—" marks cells with $k > d + 1$.

	d				
k	2	3	5	7	11
2	0.7071	0.6830	0.6545	0.6371	0.6158
3	0.8047	0.7887	0.7697	0.7581	0.7439
4	—	0.8415	0.8273	0.8186	0.8079
5	—	—	0.8618	0.8549	0.8463
6	—	—	0.8848	0.8791	0.8719
7	—	—	—	0.8963	0.8902
8	—	—	—	0.9093	0.9040
9	—	—	—	—	0.9146

7 Conclusion and Outlook

In this work we reviewed the recent literature on the joint measurability of quantum measurements and reproduced, by explicitly formulating and solving the corresponding SDP, the known noise-robustness thresholds for the joint measurability of k MUBs in prime dimension d. We then generalised this program to arbitrary compatibility hypergraphs, exposing a single framework in which full joint measurability is the trivial case. Using this generalisation we computed new noise-robustness thresholds for the all-pairs hypergraph and derived a closed-form analytical lower bound that, in every case we checked numerically, is certified by the SDP to solver precision.

Numerical contribution. Across $d \in \{2, 3, 5, 7, 11\}$ and all valid k, our SDP reproduces every closed-form entry of Designolle *et al.* [1] to within 10^{-4} (Table 1) and extends the table to the all-pairs hypergraph (Table 2), with every entry matching the analytical bound $\eta^\star = 1 - \sqrt{d}/[\,k(\sqrt{d}+1)\,]$ of (30).

Qualitative findings. The most striking qualitative finding is that the noise-robustness threshold of the pairwise hypergraph is **monotonically increasing** in k, while the corresponding threshold for full joint measurability is **monotonically decreasing** in k. The two thresholds therefore diverge as k grows: full joint measurability becomes harder, whereas pairwise compatibility becomes easier, because each measurement's "responsibility" in a pairwise decomposition is distributed across $k-1$ of the $\binom{k}{2}$ branches – a vanishing fraction $2/k$. This is the quantitative illustration, on the MUB family, of the hierarchy of compatibility hypergraphs introduced by Quintino *et al.* [5], and it sharply illustrates why genuine k-wise incompatibility is a strictly stronger notion than pairwise incompatibility.

Going further. Three directions follow naturally from this work.

(i) *A tightness certificate.* Implementing the dual SDP would either prove tightness of (30) analytically – by exhibiting a feasible dual witness whose objective matches $1 - \sqrt{d}/[\,k(\sqrt{d}+1)\,]$ – or identify the regime where the equal-weight pairwise mixture is sub-optimal.

(ii) *Intermediate hypergraphs.* Between the two extremes we studied lies a rich lattice of compatibility hypergraphs – all triples, all $(k-2)$-subsets, asymmetric structures, and so on. Our framework already handles these without further modification; only their analytical behaviour remains unexplored.

(iii) *Composite dimensions.* The Wootters–Fields construction guarantees $d+1$ MUBs only for prime-power d; in $d = 6$ no fourth MUB has ever been found, and it is widely conjectured that none exists [2]. Running the same hypergraph SDP for the three known MUBs in $d = 6$, and comparing the result to the prime-dimension pattern, would provide indirect evidence on this longstanding open question.

Acknowledgements

I am grateful to my advisors Marc-Olivier Renou and Lucas-Amadeus Tendick for their guidance throughout this project, as well as to the PHIQUS group at Inria Paris-Saclay for hosting me.

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A Appendix

A.1 Hypergraph-parameterized robustness SDP

The Python function below implements the white-noise robustness SDP generalised to an arbitrary compatibility hypergraph C, as discussed in Section 5.3. It is built on top of CVXPY and delegates variable creation, base constraints and inter-branch gluing to the helper routines `_build_variables`, `_base_constraints` and `_glue_terms` (defined in the accompanying source file `compatibility_hypergraph.py`); it returns the optimal $\eta^\star$ together with the solver's reported status.

Listing 1: `robustness_hypergraph`: noise-robustness SDP for an arbitrary compatibility hypergraph.

```python
def robustness_hypergraph(M, hypergraph, *,
                          solver="CLARABEL", **solver_kwargs) -> tuple[float, str]:
    """
    Max eta such that the noisy version of M admits a decomposition.

    Returns (eta_star, solver_status).
    """
    m, n_a, d, _ = M.shape
    I = np.eye(d)
    hypergraph = [tuple(sorted(edge)) for edge in hypergraph]
    E, J_outside, p = _build_variables(M, hypergraph)
    cons = _base_constraints(M, hypergraph, E, J_outside, p)
    eta = cp.Variable()

    for x in range(m):
        for a in range(n_a):
            noisy = eta * M[x, a] + (1 - eta) * np.trace(M[x, a]) * I / d
            cons.append(sum(_glue_terms(x, a, hypergraph, E, J_outside)) == noisy)
    prob = cp.Problem(cp.Maximize(eta), cons)
    prob.solve(solver=solver, **solver_kwargs)
    return float(eta.value), prob.status
```

Listing 2: `_glue_terms`: helper function that builds the list of contributions from each branch to $M_{x|a}$.

```python
def _glue_terms(x, a, hypergraph, E, J_outside):
    """For (x, a), build the list of contributions from each branch to M[x, a]."""
    terms = []
    for i, edge in enumerate(hypergraph):
        if x in edge:
            terms.append(_marginal_E(E[i], edge, x, a))
        else:
            terms.append(J_outside[i][x][a])
    return terms
```

A.2 The two hypergraphs we provide in the above function to generate our results

The two helper functions below construct the compatibility hypergraphs corresponding to the two extreme cases of interest. Both are intended to be passed as the `hypergraph` argument of `robustness_hypergraph` (Listing 1) together with a measurement family of k POVMs.

Listing 3: Helpers for the two extremal compatibility hypergraphs.

```python
def hypergraph_compatible(k):
    """Compatibility hypergraph for the given number of POVMs."""
    hypergraph = [tuple(range(k))]
    return hypergraph
```

```python

def hypergraph_pair(k):
    """Compatibility hypergraph with all pairs of POVMs."""
    hypergraph = [tuple(edge) for edge in itertools.combinations(range(k), 2)]
    return hypergraph
```

Full joint measurability: `hypergraph_compatible(k)`. This returns the trivial hypergraph $C = \{ (0, 1, \ldots, k-1) \}$ consisting of a single hyperedge that contains *all* k measurement settings. Plugged into `robustness_hypergraph`, the resulting SDP collapses to a single branch in which all k noisy measurements are required to share a common parent POVM $\{G_\lambda\}$ with $\lambda \in [n_a]^k$. The optimisation thus reduces exactly to the standard k-wise joint-measurability robustness program (25), whose optimal value reproduces the classical noise-robustness thresholds tabulated in Table I of [1]. This is the *tightest* constraint available—no other hypergraph yields a smaller $\eta^\star$.

All pairs: `hypergraph_pair(k)`. This returns the hypergraph $C = \{ (j, \ell) : 0 \leq j < \ell \leq k-1 \}$ consisting of all $\binom{k}{2}$ unordered pairs. Each branch i corresponds to one pair (j, ℓ) and asks for a joint POVM that reproduces only the marginals of measurements j and ℓ, while the remaining $k-2$ measurements are represented by arbitrary, independent POVMs in that branch. The SDP then searches for the maximal η such that the noisy family $\{M^\eta_{a|x}\}$ can be written as a convex (i.e. probabilistic) mixture of these pairwise-joint sub-experiments,

$$M^\eta_{a|x} = \sum_{(j,\ell)} p_{(j,\ell)} \, M_a^{(x),(j,\ell)}, \qquad p_{(j,\ell)} \geq 0, \ \sum_{(j,\ell)} p_{(j,\ell)} = 1,$$

with each branch satisfying the joint-measurability constraint only on its own pair. Operationally, the experimenter selects a pair at random with probability $p_{(j,\ell)}$ and runs the corresponding pairwise joint measurement; this exactly reproduces the observed statistics whenever the SDP is feasible. The resulting $\eta^\star$ is the white-noise threshold below which the family is *not genuinely k-wise incompatible* in the sense of [5]: pairwise joint measurability is enough to simulate the data. Since this hypergraph imposes strictly weaker constraints than full joint measurability, its threshold is monotonically larger than the one given by `hypergraph_compatible(k)` (cf. Section 5.1).